

USED CARS PRICE ANALYSIS

WEB SCRAPING STEPS

IMPORTING THE REQUIRED LIBRARIES

```
import pandas as pd
from selenium import webdriver
import time
from bs4 import BeautifulSoup
```

SCRAPING

STEP 1 : DECLARING THE URL

- url=<https://www.cars24.com/buy-used-cars-hyderabad/>
 -
 -
 -
 -
 -
 -
 -

URLS OF ALL THE CITIES

STEP 2 : AUTOMATING THE CARS24 WEBSITE TO GET SCROLLED BY USING SCROLLING LOGIC WITH SELENIUM

```
browser = webdriver.Chrome()
browser.get(url)
time.sleep(5)

while True:
    cars = browser.find_elements("css selector", "div.styles_outer__NTVth")
    if len(cars) == 0:
        break

    last_car = cars[-1]

    browser.execute_script("arguments[0].scrollIntoView({block: 'end'});", last_car)
```

```

time.sleep(3)

new_cars = browser.find_elements("css selector", "div.styles_outer__NTVth")

if len(new_cars) == len(cars):
    break
html=browser.page_source
browser.quit()

```

STEP 3 : STORING HTML CONTENT OF THE PAGE

- soup=BeautifulSoup(html,"html.parser")
- soup

STEP 4 : FINDING COUNT OF THE CARS

- cars=soup.find_all("div",class_='styles_outer__NTVth')
- len(cars)

STEP 5 : CONVERTING MESSY HTML INTO WELL FORMATTED AND HUMAN READABLE STRUCTURE USING PRETTIFY

- car=cars[0]
- print(car.prettify())

STEP 6 : CREATING EMPTY LISTS OF REQUIRED COLUMNS

```

Manufacturing_Year_list=[]
Brand_list=[]
Car_Name_list=[]
KM_Driven_list=[]
Fuel_list=[]
Transmission_list=[]
EMI_list=[]
price_list=[]

```

STEP 6 : STORING THE DATA IN EMPTY LISTS

```
for i in cars :
```

```
    name=i.find('span').get_text()
```

```
    Manufacturing_Year=name.split()[0]
```

```
    Brand=name.split()[1]
```

```
    Car_Name=" ".join(name.split()[1:])
```

```
    ul=i.find('ul')
```

```
    items=ul.find_all("p")
```

```
    KM_Driven=items[0].get_text()
```

```
    Fuel=items[1].get_text()
```

```
    Transmission=items[2].get_text()
```

```
    div=i.find('div',class_='styles_flexItem__2z__J')
```

```
    emi=div.find('span')
```

```
    if emi:
```

```
        EMI=emi.get_text()
```

```
    else:
```

```
        None
```

```
price_wrap = i.find("div", class_="styles_priceWrap__VwWBV")
```

```
new_price = None
```

```
if price_wrap:
```

```
    prices = price_wrap.find_all("p")
```

```
    if len(prices) == 1:
```

```
        new_price = prices[0].get_text(strip=True)
```

```
elif len(prices) >= 2:  
    new_price = prices[1].get_text(strip=True)
```

```
price=new_price
```

```
Manufacturing_Year_list.append(Manufacturing_Year)  
Brand_list.append(Brand)  
Car_Name_list.append(Car_Name)  
KM_Driven_list.append(KM_Driven)  
Fuel_list.append(Fuel)  
Transmission_list.append(Transmission)  
EMI_list.append(EMI)  
price_list.append(price)
```

STEP 6 : CONVERTING TO DATA FRAME

```
df=pd.DataFrame({'Car Name':Car_Name_list,  
                'Brand':Brand_list,  
                'Year of Manufacture':Manufacturing_Year_list,  
                'Distance Driven':KM_Driven_list,  
                'Fuel Type':Fuel_list,  
                'Transmission Type':Transmission_list,  
                'EMI':EMI_list,  
                'Price':price_list  
                })
```

STEP 7 : SAVING THE FILE

- `df.to_csv('cars24_hyderabad_raw.csv')`

-
-
-
-
-

SAVING ALL THE CITIES DATA

CONTINUE SAME PROCESS FOR ALL CITIES SCRAPE THEM AND SAVE